Confidence Intervals and Hypothesis Testing

Before class: complete Chapter 7 Problem 3(b)–(d). The recorded walkthrough covers part (a). Bring your reasoning, including any step you could not finish.

Goals. Identify a population parameter, construct a null reference model, choose the appropriate tail, and explain a decision in context. Work in pairs; agree on the model before reaching for a calculator.

1. A population mean and a reference claim

Reuse the illustrative data from Chapter 7 Practice 1. A laboratory randomly samples 64 independent water specimens from a large population. The mean concentration is $\overline{x} = 12.4$ mg/L and the population standard deviation is known to be $\sigma = 3.2$ mg/L. Assume the normal approximation for $\overline{X}$ is adequate. Let μ be the population mean concentration.

New question. Before sampling, the laboratory asks whether the mean concentration *differs from* the reference value 11.5 mg/L. Use $\alpha = .05$.

(a) Retrieve the Chapter 7 calculation: find $\mathrm{SE}(\overline{X})$ and the approximate 95% confidence interval for μ. Include units. Does this interval contain 11.5?

(b) State H_0 and H_a. Under H_0, write the approximate distribution of $\overline{X}$ and of

$$Z = \frac{\overline{X} - 11.5}{\sigma/\sqrt{n}}.$$

Our convention is $N(\text{mean}, \text{standard deviation})$. Calculate the observed value z and explain what it measures.

(c) Mark the observed statistic on the curve below. Shade *all* results at least as favorable to H_a. Compute the p-value, state the decision at $\alpha = .05$, and write one sentence explaining the p-value under H_0.

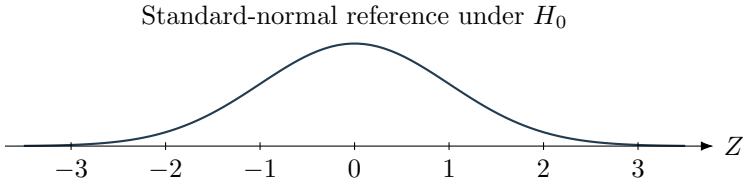


Table values: $\Phi(2.25) = .9878$, $z_{.025} = 1.96$, $z_{.005} = 2.576$.

Optional connection after the core work

Construct the 99% confidence interval. Reassess the *same* test at $\alpha = .01$. Explain which quantities change and which stay fixed. Why do the two confidence intervals agree with the corresponding two-sided test decisions here?

New testing questions extend the original Practice 1 data. The normal approximation and known σ are assumptions of this exercise.

A Sampling Probability Becomes Evidence

2. The usable-sample rate from Problem 3

Recall the illustrative monitoring-program data: 168 of 240 returned kits have usable samples. Assume independent Bernoulli outcomes with a common probability p and a population much larger than the sample. The target p is the usable-sample proportion *among this program's returned kits*.

New question. Before sampling, the program asks whether its usable-sample rate has fallen below 75%. Test the reference claim $p = .75$ using the concepts from the introductory hypothesis-testing lecture. Use $\alpha = .05$.

Recall the model from Problem 3(b). If $p = .75$, then

$$E[\widehat{p}] = .75, \qquad \mathrm{SE}_{p=.75}(\widehat{p}) = \sqrt{\frac{.75(.25)}{240}}, \qquad \frac{\widehat{p} - .75}{\sqrt{.75(.25)/240}} \overset{\text{approx}}{\sim} N(0,1).$$

This supplied reference model lets us apply the lecture's testing logic to a proportion. Check its expected success and failure counts before using the approximation.

(a) State H_0 and H_a. Which direction of deviation from .75 would favor the research question? Identify the observed $\widehat{p}$ and calculate the observed standardized statistic.

(b) On the curve, mark your observed statistic and shade the relevant tail. Explain why the probability calculated in original Problem 3(b) is now the p-value. Use the direct normal approximation, *without continuity correction*, and state your decision and contextual conclusion.

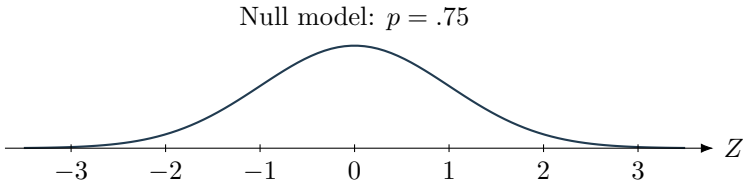


Table value: $\Phi(-1.79) = .0367$. Explain any small difference from a calculator result using an unrounded statistic.

(c) A classmate uses $\sqrt{.70(.30)/240}$ in the test because "we always plug in our sample proportion." Explain why that expression belonged to Problem 3(a), and why the null-model calculation here uses .75. In your p-value interpretation, which quantity is fixed and which varies over repeated samples?

Exit check

Keep the same data and alternative but set $\alpha = .01$. Does the p-value change? Does the decision change? Explain why failure to reject would not establish that the true rate equals .75.

Scope: hypotheses, null distributions, observed statistics, tails, p-values, and significance decisions. Critical values, rejection regions and power are developed next.